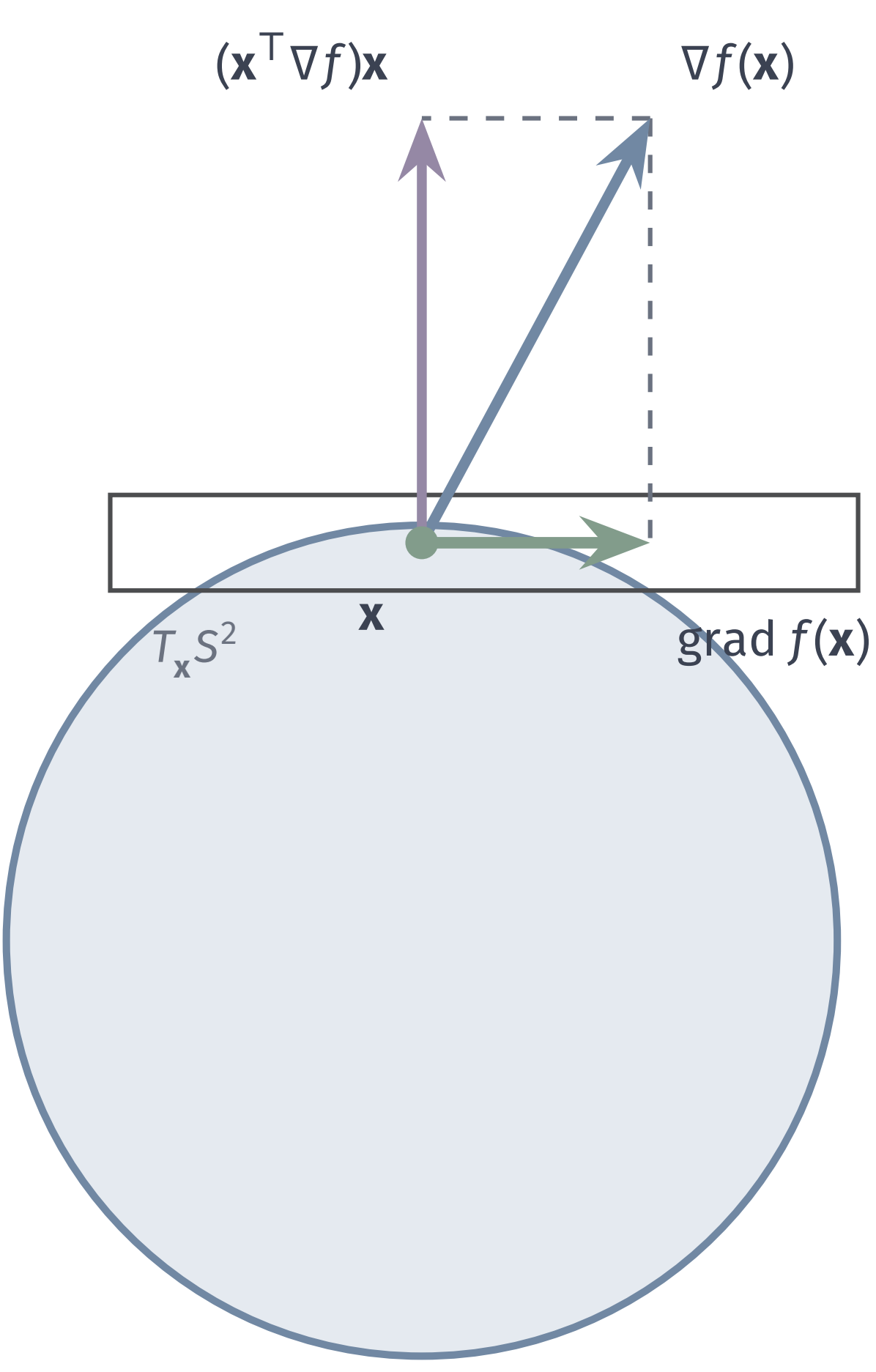
$(\mathbf{x}^\top \nabla f)\mathbf{x}$
$\nabla f(\mathbf{x})$
$\mathbf{x}$
$T_{\mathbf{x}}S^2$
grad $f(\mathbf{x})$